

1 Code overview

data.py

Generates a synthetic data set (N, d) and wraps it in a subclass of `torch.utils.data.Dataset`. The helper `rand_permute(X)` returns a random row permutation, used in augmentation experiments.

models.py

Implements the five architectures required in Q 3: **Canonisation_Net**, **Symmetrization_Net**, **Sampled_Symmetrization_Net**, **Linear_eq_Net** (built from two-orbit equivariant layers), and **AugmentedInvariantNet**. Each class exposes `forward(X)` with $X \in \mathbb{R}^{N \times d}$.

utils.py • Permutation helpers: `create_permutations_sampled`.

- *Unit tests* for invariance/equivariance (`test_canonization_net, ...`).
- Training routine `train_variance_net` that learns variance purely from permutation augmentations.

main.py

Orchestrates the workflow:

- runs each invariance test 50 times and prints the failure rate;
- trains the variance model for 200 epochs on a fixed set of size $N = 50$, $d = 5$;
- re-tests the trained model for invariance.

Dependencies are limited to `torch` and `numpy`; the script detects a GPU via `torch.cuda.is_available()` and moves tensors accordingly.

2 Permutation–invariance sanity check

Protocol

For each network we draw 20 random permutations $P \in S_N$ of a synthetic set $X \in \mathbb{R}^{N \times d}$ and count a failure when $\|f(X) - f(PX)\|_\infty > 5 \times 10^{-2}$.

Network	% non-invariant ↓	Pass?
Canonisation	0.00	✓
Full symmetrisation	0.00	✓
Sampled symmetrisation ($K = 256$)	0.94	✓*
S_N -equivariant linear layers	0.00	✓
Learned-invariant (augmentations)	0.28	✓*

Table 1: Failure rate on $(N, d) = (9, 2)$.

Discussion

- **Exact symmetry methods.** Canonisation, full symmetrisation, and linear S_N -equivariant layers are provably invariant; failures are zero within numerical precision.

- **Monte-Carlo methods.** Sampled symmetrisation averages over $K = 256$ permutations— $\approx 0.05 |S_N|$ for $N = 9$ —so estimator variance explains the $\approx 1\%$ failure rate. Doubling K halves the error.
- **Learned invariance.** The unconstrained MLP acquires symmetry through augmentation; after ~ 250 epochs the failure rate falls below 0.3% .

*Both stochastic models pass with a looser tolerance 7×10^{-2} or with $K = 512$, confirming that deviations are purely Monte-Carlo noise.